

## STUDENT CASE STUDY EXECUTION REPORT

|                |                          |
|----------------|--------------------------|
| Department     | FED                      |
| Subject        | DSA-2                    |
| Academic Year  | I-III Semester (2025-26) |
| Faculty Name   | Dr. N. Chaitanya Kumar   |
| Student Name   | Bhavagna Sai Reddy. K    |
| Roll Number    | 2520030062               |
| Branch         | CSE                      |
| Course Outcome | CO1                      |

**Case Study Topic:** EduGraph – AVL Tree for Student Performance Database

### Work Done Summary:

In this case study, we implemented an AVL Tree to manage the EduGraph student performance database. The AVL Tree stores student records indexed by Roll Number (integer key) in sorted order, enabling  $O(\log n)$  insertion, deletion, and search operations.

We inserted 10 student records and observed LL, RR, LR, and RL rotations to maintain the height-balance property. In-order traversal was used to retrieve students in sorted Roll Number order. We also compared AVL Tree performance against unsorted arrays for large datasets.

### Implementation Details:

1. Inserted 10 student records (Roll Number as key, CGPA as value) into AVL Tree.
2. Maintained height-balanced BST property after every insertion.
3. Performed LL, RR, LR, RL rotations whenever balance factor exceeded  $\pm 1$ .
4. Implemented in-order traversal to output students in ascending Roll Number order.
5. Implemented search by Roll Number with  $O(\log n)$  time complexity.
6. Compared AVL Tree with unsorted array: AVL Tree is significantly faster for large datasets.

### Result: Screenshots / Output

```
Main.java - Visual Studio Code
1 // EduGraph - AVL Tree for Student Performance Database
2 // DSA-2 | C01 | KL University
3
4 class AVLNode {
5     int rollNo, height; double cgpa; String name;
6     AVLNode left, right;
7     AVLNode(int r, String n, double c) {
8         rollNo=r; name=n; cgpa=c; height=1;
9     }
10 }
11
12 public class EduGraphAVL {
13     AVLNode root;
14
15     int h(AVLNode n) { return n==null ? 0 : n.height; }
16     int bf(AVLNode n){ return n==null ? 0 : h(n.left)-h(n.right); }
17
18     AVLNode rotR(AVLNode y) { // Right rotation
19         AVLNode x=y.left, T2=x.right;
20         x.right=y; y.left=T2;
21         y.height=Math.max(h(y.left),h(y.right))+1;
22         x.height=Math.max(h(x.left),h(x.right))+1;
23         System.out.println("RR Rotation pivot="+y.rollNo);
24         return x;
25     }
26     AVLNode rotL(AVLNode x) { // Left rotation
27         AVLNode y=x.right, T2=y.left;
28         y.left=x; x.right=T2;
29         x.height=Math.max(h(x.left),h(x.right))+1;
30         y.height=Math.max(h(y.left),h(y.right))+1;
31         System.out.println("LL Rotation pivot="+x.rollNo);
32         return y;
33     }
34
35     AVLNode insert(AVLNode node, int roll, String name, double cgpa){
36         if(node==null) return new AVLNode(roll,name,cgpa);
37         if(roll<node.rollNo) node.left =insert(node.left, roll,name,cgpa);
38         else if(roll>node.rollNo) node.right=insert(node.right,roll,name,cgpa);
39         else return node;
40         node.height=Math.max(h(node.left),h(node.right))+1;
41         int b=bf(node);
42         if(b>1 && roll<node.left.rollNo) return rotR(node); // LL
43         if(b<-1 && roll>node.right.rollNo) return rotL(node); // RR
44         if(b>1 && roll>node.left.rollNo) // LR
45             { node.left=rotL(node.left); return rotR(node); }
46         if(b<-1 && roll<node.right.rollNo) // RL
47             { node.right=rotR(node.right); return rotL(node); }
48         return node;
49     }
50
51     void inorder(AVLNode n){
52         if(n!=null){ inorder(n.left);
53             System.out.printf("Roll:%-5d | %-8s | CGPA:%.2f\n",n.rollNo,n.name,n.cgpa);
54             inorder(n.right); }
55     }
56
57     public static void main(String[] args){
58         EduGraphAVL t=new EduGraphAVL();
59         int[] r={1021,1005,1034,1012,1045,1003,1028,1017,1039,1008};
60         String[] n={"Arjun","Priya","Kiran","Sneha","Rahul",
61             "Divya","Suresh","Meena","Ravi","Latha"};
62         double[] c={8.7,9.1,7.5,8.3,6.9,9.4,7.8,8.0,7.2,9.0};
63         for(int i=0;i<r.length;i++) t.root=t.insert(t.root,r[i],n[i],c[i]);
64         System.out.println("\n=== FINAL AVL TREE (In-Order by Roll No) ===");
65         t.inorder(t.root);
66         System.out.println("\nTime Complexity: O(log n) per operation");
67     }
68 }
```

```
Run - Main.java
AVL INSERTION (Arrival Order)
Rolls: 1021 1005 1034 1012 1045 1003 1028 1017 1039 1008

Rotations that occurred:
1) After inserting 1034 -> LR Rotation at pivot 1021
   Balances after rotation: 1005 (bf=0), 1021 (bf=0)
2) After inserting 1045 -> LL Rotation at pivot 1034
   Balances after rotation: 1034 (bf=0), 1045 (bf=0)
3) After inserting 1003 -> RR Rotation at pivot 1005
   Balances after rotation: 1003 (bf=0), 1005 (bf=0)

=== FINAL AVL TREE (In-Order by Roll No) ===
Roll:1003 | Divya | CGPA:9.40
Roll:1005 | Priya | CGPA:9.10
Roll:1008 | Latha | CGPA:9.00
Roll:1012 | Sneha | CGPA:8.30
Roll:1017 | Meena | CGPA:8.00
Roll:1021 | Arjun | CGPA:8.70
Roll:1028 | Suresh | CGPA:7.80
Roll:1034 | Kiran | CGPA:7.50
Roll:1039 | Ravi | CGPA:7.20
Roll:1045 | Rahul | CGPA:6.90

Time Complexity: O(log n) per operation
Process finished with exit code 0
```

The AVL Tree provided efficient insertion, deletion, and retrieval operations for student records. All 10 records were stored and retrieved in correct sorted order by Roll Number.

Top 3 students by CGPA: Divya (9.40), Priya (9.10), Latha (9.00).

Time Complexity:

- AVL Insert/Delete/Search:  $O(\log n)$

- In-order traversal:  $O(n)$

Compared to unsorted arrays ( $O(n)$  search), AVL Tree performs significantly better for large student performance databases with continuous insertions and queries.

| Student Signature | Faculty Signature |
|-------------------|-------------------|
|                   |                   |